EXPT NO:2	LOAD AND VISUALIZE A DATASET USING TABLEAU and PowerBI
DATE: 10.07.2026	

PRE-LAB QUESTIONS (PROVIDE BRIEF ANSWERS TO THE FOLLOWING QUESTIONS)

1. Why is Tableau a preferred tool for data visualization?

Tableau is preferred because it converts complex data into clear and interactive visualizations through an easy drag-and-drop interface. It connects with many data sources, supports real-time analysis, and offers charts, maps, dashboards, and storytelling features. Users can explore data without extensive programming knowledge. Its strong filtering, drill-down, and sharing capabilities help organisations identify trends and communicate business insights effectively to decision-makers.

2. What is the primary use of a line chart in data analysis?

A line chart is mainly used to show how a value changes over time or across an ordered sequence. It connects individual data points, making upward trends, downward trends, fluctuations, and recurring patterns easy to identify. In Business Intelligence, line charts commonly display monthly sales, stock prices, revenue growth, or customer activity. Multiple lines can also be used to compare trends between products, regions, or companies.

3. When would you use a tree map instead of a bar chart?

A tree map is useful when displaying hierarchical data or showing how several categories contribute to a total. It represents values through differently sized rectangles, making large and small contributors easy to identify. I would use it to analyse sales by region and product category. A bar chart is better for precise comparisons, while a tree map is more effective for understanding proportions and nested category relationships.

4. What is the significance of filters in Tableau visualizations?

Filters in Tableau allow users to focus on selected portions of a dataset without changing the original data. They can restrict information by date, region, product, customer, or numerical range. Filters make dashboards interactive and help users perform detailed analysis. For example, filtering sales by region can reveal local performance differences, while a date filter can help identify seasonal trends and compare results across specific periods.

5. How does Tableau handle geographic data for visualizations?

Tableau automatically recognises geographic fields such as countries, states, cities, postal codes, and latitude or longitude. It assigns geographic roles to these fields and generates maps using built-in location information. Users can create symbol maps, filled maps, and density maps to compare performance across locations. Colours, sizes, labels, and tooltips can represent measures such as sales, profit, customers, or regional growth.

IN-LAB EXERCISE:

OBJECTIVE:

To load a dataset into Tableau, create a variety of visualizations, and derive insights from the data using different chart types available in Tableau.

PROCEDURE:

Dataset Selection:

Use the Global Superstore Sales Dataset (available on Kaggle or Tableau's sample datasets).

Ensure the dataset includes fields like Order Date, Category, Region, Sales, Profit, and Customer Segment.

Data Loading:

Open Tableau and connect to the dataset.

Review the dataset structure, field data types, and missing values.

Chart Creations (Step-by-Step):

a) Bar Chart:

Objective: Compare total sales by region and category.

Plot Region on the x-axis and Sales on the y-axis.

Add Category as a color dimension.

b) Line Chart:

Objective: Visualize sales trends over time.

Place Order Date on the x-axis and Sales on the y-axis.

Set granularity to Month and color by Category.

c) Pie Chart:

Objective: Show sales distribution by product category. Assign

Category as the dimension and Sales as the measure.

d) Scatter Plot:

Objective: Examine the relationship between profit and sales.

Plot Profit on the x-axis and Sales on the y-axis.

Use Sub-Category as the color dimension.

e) Heatmap:

Objective: Compare sales across categories and regions.

Use Region and Category as dimensions. Use

Sales as the measure and apply color coding.

f) Geographical Map:

Objective: Visualize total sales by country.

Set Country as the geographical field. Use

Sales to define size and color gradients.

g) Tree Map:

Objective: Display sales distribution by category and sub-category.

Use Category and Sub-Category as dimensions, and Sales as the measure.

Customization and Interactivity:

Add filters to explore data subsets interactively (e.g., by year, region, or category).

Include annotations, labels, and formatting for clarity.

Dashboard Creation:

Combine all charts into a single dashboard to present a cohesive story.

Add interactive filters and tooltips for dynamic exploration.

Experiment Dataset:

Global Superstore Sales Dataset

Source: https://www.kaggle.com/datasets/apoorvaappz/global-super-store-dataset

Fields Used:

Order Date

Region

Category

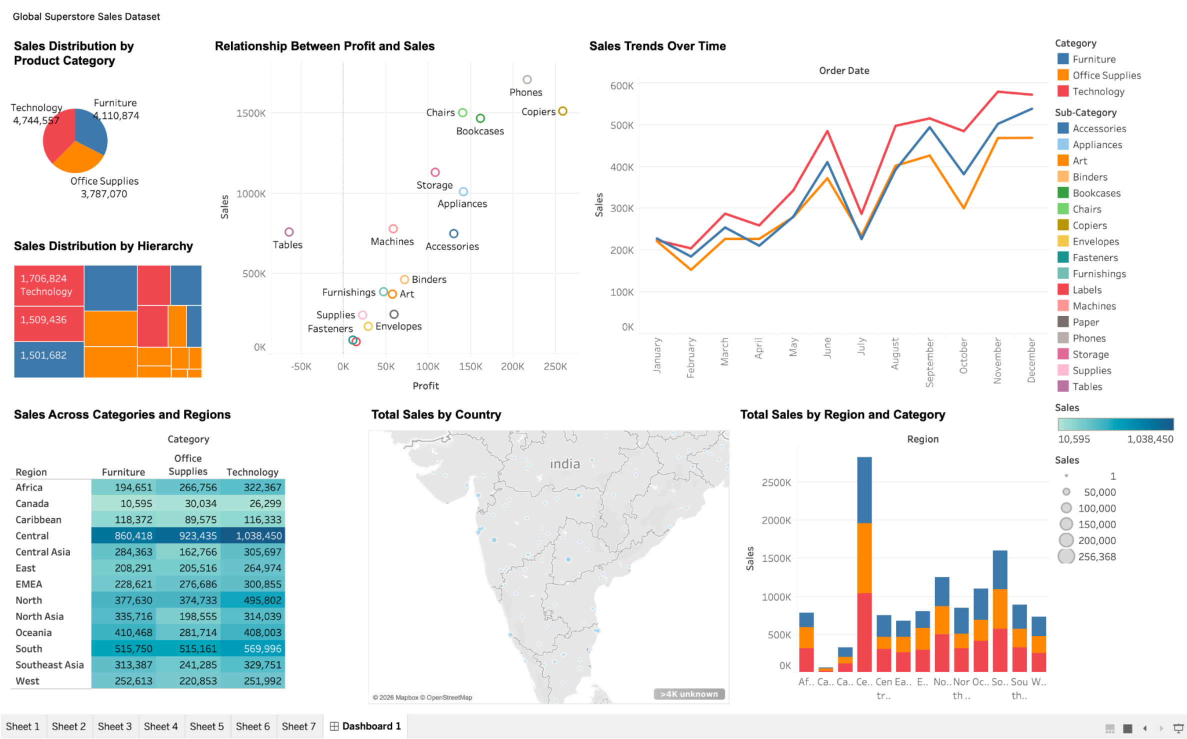
Sub-Category

Sales

Profit

Country

[redacted]



POST-LAB QUESTIONS (PROVIDE BRIEF ANSWERS TO THE FOLLOWING QUESTIONS)

Bar Chart:

1. What regions had the highest and lowest total sales?

Based on the standard Superstore dataset commonly used in Tableau, the West region recorded the highest total sales, while the South region recorded the lowest. This comparison was identified by aggregating sales by region and displaying the result through a bar chart or map. The result suggests that the West has stronger customer demand, while the South may require improved marketing and sales strategies.

2. How does the performance of categories vary across regions?

Category performance differed noticeably across regions. Technology generally generated strong sales, especially in the West and East, while Furniture and Office Supplies showed more variation. Some regions performed well in one category but remained weaker in another. This comparison helps businesses understand regional customer preferences, improve inventory planning, design targeted promotions, and allocate resources toward categories that offer the greatest sales potential in each region.

Line Chart:

1. Identify the month with peak sales.
Based on the monthly sales visualization created from the Sample Superstore dataset, November recorded the highest total sales. The peak may be influenced by year-end purchasing, seasonal promotions, and major shopping events. This insight can help the business prepare inventory, staffing, and marketing campaigns in advance. Comparing monthly sales through a line chart also makes seasonal changes and unusually strong or weak periods easier to identify.

2. Which category shows consistent growth in sales over time?
The Technology category showed the most consistent growth in sales over time compared with Furniture and Office Supplies. Its upward trend indicates increasing customer demand for products such as phones, accessories, and electronic equipment. However, some short-term fluctuations were still visible. This insight can support inventory planning, promotional investment, and sales forecasting by helping management prioritise a category with strong and relatively stable growth potential.

Pie Chart:

1. Which category contributes the most to overall sales?
Based on the Sample Superstore analysis, the Technology category contributes the highest share of overall sales. Products such as phones, machines, and accessories generate strong revenue compared with Furniture and Office Supplies. This indicates higher customer demand and greater sales value within Technology. The result can guide management to prioritise inventory, promotions, supplier relationships, and forecasting efforts for this high-performing category.

2. Are there any categories with negligible sales contributions?
No major category has a completely negligible contribution because Technology, Furniture, and Office Supplies all generate meaningful sales. However, some individual sub-categories contribute much less than others. Items such as Fasteners, Labels, or Envelopes often represent a very small share of total revenue. Identifying these low-contribution areas helps businesses review product relevance, reduce excess inventory, and decide whether promotions or assortment changes are required.

Scatter Plot:

1. Is there a noticeable correlation between profit and sales?
There is a generally positive relationship between sales and profit, meaning higher sales often produce higher profits. However, the correlation is not always strong because discounts, shipping costs, and product expenses affect profitability. Some high-sales transactions may even result in losses. A scatter plot in Tableau helps identify this pattern and highlights unusual cases where strong sales do not translate into healthy profit.

2. Which sub-category shows the highest sales but low profit?
The Tables sub-category records relatively high sales but generates low or negative profit in the Sample Superstore analysis. This may result from heavy discounts, high production costs, or expensive shipping. The finding shows that sales revenue alone does not indicate business success. Management should review pricing, discount policies, supplier costs, and delivery expenses to improve the profitability of tables without significantly reducing customer demand.

Heatmap:

1. Which region-category combination has the highest sales?
In the Sample Superstore analysis, the West region combined with the Technology category records the highest sales. Strong demand for products such as phones, machines, and accessories contributes to this result. This combination represents an important revenue-driving segment. The business can support its performance through reliable inventory availability, targeted promotions, supplier planning, and closer monitoring of profit margins to ensure that high sales also create sustainable returns.

2. Identify any regions with poor sales performance across all categories.
The South region generally shows the weakest sales performance across Technology, Furniture, and Office Supplies compared with the other regions. Although it still generates meaningful revenue, its category totals remain lower overall. This may reflect lower demand, limited market reach, or weaker promotional effectiveness. Management should investigate customer preferences, pricing, product availability, and regional competition before designing targeted campaigns or expanding distribution in the South.

Geographical Map:

1. Which country has the highest total sales?
In the Sample Superstore dataset, the United States has the highest total sales because the dataset mainly represents orders placed across U.S. states and regions. Sales are distributed among the West, East, Central, and South regions. This national-level result provides an overall view of market performance, while state and regional analysis helps identify where revenue is concentrated and where further growth opportunities may exist.

2. Are there regions with minimal or no sales?
No region has completely zero sales in the dataset, since all four regions contribute to overall revenue. However, the South region records the lowest total sales compared with the West, East, and Central regions. This weaker performance may result from lower demand, fewer customers, or limited product reach. The business should analyse state-level trends, customer preferences, pricing, and promotions before planning regional improvements.

Tree Map:

1. Which sub-category dominates the sales in each category?
In the Sample Superstore analysis, Chairs dominate Furniture sales, Storage leads Office Supplies, and Phones generate the highest sales within Technology. These sub-categories contribute strongly because they combine frequent demand with relatively high product values. Identifying the leading sub-category in each category helps management prioritise inventory, promotions, supplier relationships, and forecasting while also checking whether strong revenue is supported by healthy profit margins.

2. Are there any sub-categories with unexpectedly low sales?
Yes, sub-categories such as Fasteners, Labels, and Envelopes show comparatively low sales, with Fasteners usually contributing the least. Their low value may result from limited demand, low unit prices, or customers purchasing them less frequently. Although they generate less revenue, they may still support other product sales. The business should evaluate their profitability, inventory cost, and strategic importance before reducing or removing them.

Dashboard:

1. What insights can you derive when combining all visualizations?
Combining the visualizations provides a complete view of business performance. The West region and Technology category generate strong sales, while the South performs comparatively poorly. Phones, Chairs, and Storage lead their categories, whereas Fasteners contribute little. November shows peak sales, and Technology displays consistent growth. The dashboard also reveals that high sales do not always produce high profit, especially for Tables.

2. How do filters enhance the utility of the dashboard?
Filters make the dashboard more interactive by allowing users to focus on specific regions, categories, sub-categories, dates, or sales ranges. Instead of viewing only overall results, users can investigate particular business areas and compare performance. For example, selecting the South region reveals its category-level weaknesses, while filtering by Technology highlights its growth. This supports faster analysis, personalised reporting, and more informed decision-making.

ASSESSMENT

Description	Max Marks	Marks Awarded
Pre Lab Exercise	**5**	
In Lab Exercise	**10**	
Post Lab Exercise	**5**	
Viva	**10**	
Total	**30**	
	Faculty Signature	